

---

# Development guidelines applicable for Qualcomm 6690 (Dragonwing) Devices

TC501, TC701, TC201 & ET4 Tablets.

---

# Flexible Connection Modes

## RFID\_ONLY Mode (Max Performance)

- **How it works:** Dedicates the device's full resources to the RFID reader while temporarily pausing cellular connectivity.
- **Benefit:** Eliminates potential radio interference to guarantee the fastest read rates and peak throughput.

## RFID+WAN Mode (Real-Time Sync)

- **How it works:** Allows standard data operations to co-exist simultaneously with active RFID scanning.
- **Benefit:** Enables instant data updates while keeping cellular data active. It intelligently prioritizes voice calls, pausing RFID scanning only while you are on a call.

## Seamless API Control:

- By default, the device starts in **RFID\_ONLY** mode to ensure the best possible scanning performance right away.
- Developers can easily switch the device to RFID+WAN mode using this simple code command:  
`void connect(ENUM_CONNECTION_MODE mode)`

# WWAN + RFID use cases: SIM combination

| SIM 1 Status                    | SIM 2 (eSIM) Status             | Resulting System Behavior                                                                        |
|---------------------------------|---------------------------------|--------------------------------------------------------------------------------------------------|
| Configured                      | Not Configured                  | RFID and VoIP calls can co-exist, while cellular calls are paused during RFID use on SIM 1.      |
| Not Configured                  | Configured                      | RFID and VoIP calls can co-exist, while cellular calls are paused during RFID use on SIM 2.      |
| Configured with data preference | Configured                      | RFID, Data apps, and VoIP apps co-exist on SIM 1. SIM 2 is paused to ensure a stable connection. |
| Configured                      | Configured with data preference | RFID, Data apps, and VoIP apps co-exist on SIM 2. SIM 1 is paused to ensure a stable connection. |

# Performance Management

| Cellular (WAN) State | Resulting RFID Behavior                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------|
| Idle                 | ✅ Maximum Performance: Delivers full RFID throughput for high-speed scanning.                              |
| Hold Voice/Data      | 🧠 Balanced Performance: Enables simultaneous scanning and data use by expertly sharing resources.          |
| Active Voice Call    | 📞 Prioritizes Voice: Intelligently pauses RFID to ensure perfect call clarity, then automatically resumes. |

---

# Built-in Safeguards for Reliability & Safety

- **Stable USB Connection:** To ensure a stable data connection and prevent conflicts, RFID is not active while the USB cable is plugged in for charging or data transfer.
- **Low Battery Protection:** To preserve power for critical functions, the RFID reader automatically enters a low-power state when the battery is below 10%.
- **Airplane Mode:** As required by aviation regulations, all radios, including RFID, are disabled in Airplane Mode.
- **Idle Power-Save:** To maximize battery life, the RFID connection automatically sleeps if idle for more than 9 minutes and instantly wakes when you resume scanning.
- **Prioritizing Communications:** The system always prioritizes voice calls. RFID is temporarily paused during a VoIP call to ensure clear communication and will resume after.

---

# Single-Client Architecture for Maximum Stability

- **Exclusive Reader Control:** Only one application has control of the RFID reader at any given moment. This intentional safeguard prevents background programs from interrupting active scans or sending conflicting commands.

*This is why you cannot run both Data Wedge and 123RFID at the same time, ensuring your active application has uninterrupted access to the scanner.*

# Feature Updates & New Architectural Approach

| Features Not Supported                                              | Architectural Approach & Benefit                                                                                           |
|---------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| AB Flip Configuration (This is planned to be supported in future) ) | Replaced by high-performance session management directly on the chip, optimized for faster inventory workflows.            |
| Separate RFID Firmware Updates                                      | RFID enhancements are now part of standard OS updates, simplifying IT maintenance and ensuring system stability.           |
| Periodic Start/Stop Triggers                                        | Eliminates background battery waste by waking up instantly only when the scan button is pressed.                           |
| Custom (CSP)                                                        | CSP is focused strictly on foundational needs (factory reset, region configuration) for a more secure and stable platform. |
| Data wedge don't support RFID only mode                             | By default, Data wedge connects to RFID + WAN mode to support simultaneous tag scanning and cellular connectivity          |

---

# Frequently Asked Questions and Answers

**Q: Is RFID enabled when in an eConnex cradle?**

**A:** Yes, it is enabled. However, it is automatically disabled under two specific conditions

- When the device is actively charging.
- When an external RFID reader is attached to the cradle.

**Q: Does the "Low Battery" state affect transmit power or duty cycle?**

**A:** It directly reduces transmit power to conserve the remaining battery life.

**Q: Does "Idle Power" mode impact all device functions?**

**A:** No, it exclusively applies to inventory rounds, saving power without disrupting other system tasks.

**Q: What regions are supported on Qualcomm 6690 powered devices?**

**A:** The complete list is available on [\[SharePoint link\]](#)



---

# Recommended settings by use cases

- Inventory operations in small tags population  
[ session: S0 or S1, link: PR-ASK M4 40KHz or PR-ASK M2 80KHz, Tx power 25 dBm]
- Inventory operations in large tags population  
[ session: S2, S3, link: PR-ASK M1 160KHz or PR-ASK M2 80KHz, Tx power 25 dBm]
- Read/write operations on specific tags  
[ session: S0, link: PR-ASK M4 40KHz or PR-ASK M4 40KHz, reduce Tx power]